INCH-POUND

MIL-T-14521E
5 October 1990
SUPERSEDING
MIL-T-14521D
16 May 1985

MILITARY SPECIFICATION

TRESTLES. MOTOR VEHICLE MAINTENANCE AND STORAGE

(VEHICLE SUPPORT STANDS)

this specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers adjustable trestles (vehicle support stands), of the rack and pawl types, positioned under raised motor vehicle to support the vehicle at predetermined heights during maintenance or storage. Trestles are designed to be used in pairs. The terms trestle and vehicle support stand are used interchangeably in this specification.

1.2 Classification. Trestles covered by this specification shall be of the following load capacities, as specified (see 6.2.1).

2 US Tons
5 US Tons
7 US Tons
10 US Tons

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation (see 6.2.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4910

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SPECIFICATIONS

FEDERAL

PPP-B-621 - Boxes, Wood, Nailed and Loci-Carner.

MILITARY

MIL-P-116 Preservation, Methods of.
MIL-P-19834 - Plates, Identification or Instructor,. Metal Foil, , Adhesive Backed, General Specification for.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-129 - Marking 'or Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals,
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking. and Waterproofing: with Appropriate Test Methods.
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing, and Marking,

(Copies of specifications, standards, and other Government drawings and publications required by contractor in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting acitivity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2.1).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E4 - Load Verification of Testing Machines, Methods of,

(Application for copies should be addressed to the American Society for Testing and Materials. 1916 Race St.. Philadelphia, PA 19103.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

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ASME/ASNI PALD-4 - Vehicle Support Stands.

(Application for copies should be addressed to the American Society for Mechanical Engineers, 345 East 45th Street. New York, NY 10017.

(Non-Government standards and other publications are normally available from the organizations which prepare or which distributed the documents. These documents also may be available in or through libraries or other Informational services.)

2.3. Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets, or MS standards) , the text of this document tales precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained,

3. REQUIREMENTS

3.1 First Article. the specified (see 6.2.1), a sample shall be subject to first article inspection (see 6.3) in accordance with 4.2.1. The approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the supplier of the responsibility to furnish equipment in accordance with the requirements of the specification.

3.2 Illustrations. Figures 1, 2, 3, and 4 contain illustrations that are intended to be descriptive, not restrictive, and are net intended to preclude the procurement of trestles otherwise conforming to this specification.

3.3 Design. The trestles shall conform to the requirements of table I. The trestles shall be new and of the rack and pawl type only. The trestles shall be designed for positioning under raised motor vehicles and shall be capable of supporting the specified load at various heights. The trestles shall meet ail of the requirements of ASME/ANSI PALD-4 and all of the requirements of this specification.

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TABLE I. Trestle dimensions.

Load Capacity (US Tons)	Maximum Height Closed (Inches)	Minimum Height Extended (Inches)	Saddle Width (Inches)	Saddle Outside Horizontal Length (Inches) (See D, Figure 2)
2	12	17	Not less than 1.0	Not less than 5.0
5	16	24	From 1.31 to 1.62	Not less than 5.6
7	21	30	From 1.31 to 1.62	Not less than 5.8
10	19	30	Not less than 1.62	Not less than 6.1

3.4 Construction. The trestles shall be constructed of parts and materials free from defects that will prevent the trestles from meeting any of the requirements of this specification.

3.4.1 Materials. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the trestle manufacturing industry, shall be suitable for the intended purpose in the design of the trestle and shall be free from defects which would adversely affect the performance of the assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.4.2 Reclaimed materials. The manufacturer may use reclaimed materials for fabricating new parts. Reclaimed materials shall be reprocessed. remanufactured, or recycled in a manner that restores them to the same chemical composition and physical properties as the materials originally selected for use. Use of reclaimed parts as is or rebuilt from scrap or other used equipment shall not be permitted.

3.4.3 Castings and forgings. Castings and forgings shall be free of defects which affect serviceanility or performance, fins, scale, inclusions, cold shuts, voids, cracks, thermal ruptures, laps, folds, and mismatching. Defective castings and forgings shall not be reclaimed for use on this end item.

3.4.4 Fastening devices. Fastening devices and methods shall be chosen to serve the need while providing necessary adjustablility for service, maintenance or repair, Fasteners shall be installed to prevent loss of tightness and shall not loosen in service.

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7.4.4.1 Welding. Welding shall be neat in appearance and shall be strong enough to withstand application of the proof load and off-center rated loads to the completed trestle without cracking, or other damage.

3.4.5 Painting. All surfaces of the trestles. inside and out, shall be painted in accordance with the best practice of the commercials jack and vehicle support stand industry. Dried paint shall net interfere with the free movement of the pawl.

3.5 Product performance characteristics. The trestles shall be capable of supporting the proof load and off-center rated load in all height positions from fully lowered to fully extended with no failure or permanent deformation exceeding that specified in 3.5.4.

3.5.1 Preload. At the contractors option. a preload not exceeding fifty percent of the rated load may be applied as depicted in Figure 1 , to settle the trestle in its load carrying configuration, remove looseness and allow for minor permanent set and peening of metal parts. The preload, if elected. shall be applied prior to any measurements or testing. The preload shall be applied only once at the fully extended position and only once at the fully lowered position and nowhere in between.

3.5.2 Off-center load. The trestles shall not become unstable when subjected to the rated load capacity, applied by a universal compression testing machine, on one lip of the saddle and then on the other lip (see Figure 3) at any and all positions from fully lowered to fully extended. The trestles shall evidence no sign of column bending, visible cracks, component failure lure or permanent deformation exceeding that specified in 3.5.4 after being loaded on each lip of the saddle.

3.5.3 Proof load. The proof load shall be 1.5 times the rated load capacity. With the proof load applied by a universal compression testing machine on both lips of the saddle (see Figure 4) , the trestle shall sustain the proof load at any and all positions from fully lowered to fully extended, with no permanent deformation exceeding that specified in 3.5.4. no cracked welds, no mechanical failure, and no component damage.

3.5.4 Deformation. Permanent deformation of the trestles after any and all load tests shall not exceed 0.063 inches in the trestle base length and saddle height, dimensions A and B of Figure 2. when compared with the base line measurements.

3.6 Component parts. The trestles shall consist of a base, a one piece column and saddle with rack, a pawl, a handle and any other components needed to effect an adequate design for the intended purpose.

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3.6.1 Base. The trestle base shall have four legs forming a pyramid configuration symmetrical about the column. The trestle shall be supported by the bottom of the legs only. No pads or supporting plates shall be used on the bottom of the legs. The length and width of the base shall meet the requirements of ASME/ANSI PALD-4. The base shall afford a sturdy structure for the distribution of the supported load. The upper portion of the base, called the collar, shall position and retain the column and saddle and shall not deform under proof load or off-center loading.

3.6.2 Column and saddle. The column and saddle shall support the load and transfer teh force through the collar to the trestle base. The saddle is the load contacting and bearing part of the column and shall provide safe retention of frame members, under carriage, lower control arms, cross members, spring mountings and shackles, axle and axle housings of vehicles. The saddle height shall be adjustable in increments of not more than 1.5 inches from fully lowered to fully extended. Retention lips at the ends of the saddle shall be at least 0.25 inch high on the inside dimension. The column and saddle shall be integrated as a single piece casting or forging and shall have the rack machined, cast, or forged into the column, The rack shall meet the requirements of 3.6.3. The trestle shall be designed such that the bottom end of the column shall not come closer than 0.25 inch to the tester platen when subjected to the proof load with the column in the lowest possible position. Outside saddle dimensions shall be as specified in Table i. The design of the trestle shall prevent separation of the column and saddle from the base when the trestle is lifted by the column or saddle.

3.6.3 Rack and pawl. The configuration of the rack and pawl shall be such that the pawl engages the rack at every tooth by action of gravity alone as the rack is raised past the pawl. The pawl shall be integral with the handle and shall utilize the weight of the handle to engage the pawl with the teeth of the rat} . The trestle shall contain no springs or other devices to cause the pawl to engage the rack , or to augment the force imparted by gravity. Deliberate action by the operator shall be required for rack and pawl disengagement, in addition. the teeth of the rack shall be designed to interlock with the pawl as a load is applied from above, so that the pawl cannot be disengaged from the rack without removing the load and raising the saddle slightly. The wear of the rack and pawl shall be self compensating.

3.6.4 Handle. A handle shall be provided that shall accommodate one-handed lifting and carrying of the trestle for movement around a shop.

3.7 Identification marking. Trestles shall be permanently marked in accordance with MIL-STD-130, including the National Stock Number, contract number , manufacturer 's name.. nomenclature and load capacity. In addition, the rated capacity in tons shall be cast or stamped into both the trestle column and base in characters no less than .5 inch high.

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3.7.1 Caution plate. A caution plate sitting no less than the following shall be applied to the trestle on the same side as the handle.

CAUTION
DO NOT OVERLOAD STANDS.
PLACE LOAD ON CENTER OF SADDLE ONLY.
USE STANDS IN PAIRS, ON HARD LEVEL SURFACE.
LOAD AND STANDS SHALL BE STABLE.
STUDY , UNDERSTAND, AND FOLLOW ALL INSTRUCTIONS.
FAILURE TO HEED THIS WARNING MAY RESULT
IN PERSONAL INJURY AND/OR PROPERTY DAMAGE.
MANUALLY ENGAE LOCKING PAWL PRIOR TO USE

The caution plate shall be fabricated of adhesive backed aluminum foil plate in accordance with MIL-P-19634 or shall be metal plate, permanently fastened and permanently marked.

3.8 Workmanship. Workmanship of the trestles and accessories shall be the best quality prevailing among manufacturers commercially Producing and marketing trestles of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any ether facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification whet-e such inspections are deemed necessary to ensure supplies and servicecs conform to the prescribed requirements.

4.1.1 Responsiblity for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material , either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1)
(b) Quality conformance inspection (see 4.2.2)

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4.2.1 First article inspection. When specified (see 6.2.1), first article inspection shall be performed on any preproduction model (s) or initial production item (s) noted in 3.1. Unless otherwise specified (see 5.2.1) , first article inspection shall consist of the examination in 4.3, and the tests in 4.4, Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.2.2 Quality conformance inspection.

4.2.2.1 Inspections and tests. Quality conformance inspection shall be applied to production units offered for acceptance under the contract. Quality conformance inspection shall consist of (a) through (c) below. Failure of any unit to pass an examination or test shall be cause for rejection of that unit.

(a) Product examination (see 4.3)
(b) Production test and test methods (see 4.5)
(c) Packaging inspection (see 4.6)

4.2.2.2 Sampling. Sampling for quality conformance inspection shall be performed as listed with all samples selected randomly

INSPECTION OR TEST	LOTS OR BATCH SIZE	SAMPLE SIZE
Product examination (4.3)	1-150	1
Off-center loading test - production (4.5.1) 1!	1-150	13
Proof load test - production (4.5.2)	1-150	100%
Functional test - production (4.5.3)	1-150	13
Packaging inspection (4.6)	1-150	5

The lots shall not exceed the maximum sizes indicated above. If lot size is less than or equal to sample size. 100 percent inspection is required. Each lot shall be accepted with no defects and rejected if one or more defects are found.

4.3 Product examination. Visually, dimensionally. and manually examine each trestle to determine conformance with the requirements of 3.3 through 3.4.5 and 3.6 through 3.8. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleanign, and freedom from identified defects. Dimensional examination includes measuring dimensions including weights, as specified. Manual examinations shall include the operation of movable parts by

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hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the trestle has passed the examination prescribed for it by the contractor's iris.pectlcn system shall be cause for rejection.

4.4 First Article test and test methods. The baseline measurement and tests of 4.4.3. 4.4.4, and 4.4.5 shall be performed in the order given. and in accordance with the requirements of 4.4. 1 and 4.4.2.

4.4.1 Testing machine. The test specified in 4,4.4, 4.4.5, 4.5.1. and 4.5.2 shall be conducted using a compression testing machine which is accurate within five percent of the load being measured. The accuracy of the testing machine, including the effects of off-center (eccentric) loading, shall be verified in accordance with ASTM E4. Dead loads are considered unsafe for test purposes and shall 1 not be used. All loads shall be applied through platens of such sufficient thickness and hardness that they will not deform or coin under the loads applied to the trestle. The platens shall be flat and smooth and shall not restrain the trestle from tipping, moving, or deforming during load testing.

4.4.2 Examination and measurements. All measurements shall be performed while the trestle is resting on a surface which is flat and smooth within 0.001 inch. If suitable. the lower platen of the testing machine may be used for this purpose. The measuring device used shall be accurate within 0.0005 inch. The trestle shall be visually examined for cracked welds and other damage. Following the examination, the trestle base length and width and the distance from the surface plate or lower platen to the top of each lip of the saddle shall be measured (see dimensin B of Figure 2) with the saddle and column fully raised. Also, the distance between the bottom end of the column and the platen shall be measured with the column in the lowest possible position (see 3.6.2).

4.4.3 Baseline measurements. Friar to testing, but after the preload if used (see 3.5.1). the trestle shall be examined and baseline measurements shall be taken in accordance with 4.4.2. The baseline measurements shall be performed only once for each trestle. The baseline shall be used for comparison with similar measurements taken after the tests of 4.4.4 and 4.4.5 to determine the extent of permanent deformation of the trestle, if any. Evidence of cracked welds or other damage shall be cause for rejection of the trestle (see 3.5.1).

4.4.4 Off-center loading tests. The following test shall be performed on each end. in turn, of the trestle saddle. The trestle shall be positioned in the test machine with the saddle and column fully extended. A load equal to the trestle's specified capacity shall be applied to the saddle for a time period of one hour, as depicted in Figure 3. The load shall then be removed

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and the trestle examined for damage and for permanent deformation in accordance With 4.4.2. Following the examination and measurements. the operability of the rat} and pawl mechanism shall be verified. Evidence of cracked welds or other damage: permanent deformation in excess of the limits specified (see 3.5.4 and 3.6.2): failure of the rack and pawl mechanism to operate freely throughout it's entire adjustment range: or any tipping, overturning, or movement of the trestle shall be cause for rejection (see 3.5.2). The above test shall be repeated with the saddle and column of the trestle fully retracted (a total of four load tests shall be performed.)

4.4.5 Proof load test. The trestle shall be positioned in the testing machine with the saddle and column fully extended. A proof load equal to 1.5 times the trestle's specified capacity shall be applied to the saddle for a time period of one hour, as depicted in Figure 4. The proof load shall then be removed and the trestle examined for damage and measured for permanent deformation in accordance with 4.4.2. Following the examination and measurement, the operability of the rat} and pawl mechanism shall be checked. Evidence of cracked welds or ether damage: Permanent deformation in excess of the limits specified (see 3.3.4 and 3. 6.2): or failure of the rack and pawl mechanism to operate freely throughout it's entire adjustment range shall be cause for rejection (see 3.5.3). The above test shall be repeated with saddle and column fully retracted.

4.4.6 Functional test. After the completed trestles have been painted and the paint has dried, the trestles shall be tested for proper operation of the rack and pawl mechanism. The test shall consist of raising and lowering the saddle by hand. allowing the pawl to engage each tooth of the rack in turn. At each increment of saddle height adjustment. an attempt shall be made to dislodge the pawl by pushing on, shaking, and turning the saddle. When the saddle is fully extended, the trestle shall be lifted by the saddle. Failure of the pawl to automatically engage each tooth of the rack as the saddle is lifted: ability to dislodge the pawl from the rack through any movement of the saddle other than straight up: separation of the column and saddle from the base when the trestle is lifted by the saddle; or inability to lower the raised saddle shall be cause for rejection (see 3.6.2 and 3.6.3).

4.5 Production test and test methods. The tests 4.5.1, 4.5.2, and 4.5.3 shall be performed in the order given and in accordance with the sampling plan. Tests 4.5.1 and 4.5.2 shall be performed in accordance with the requirements of 4.4.1.

4.5.1 Off-center loading test-production. The following test shall be performed on each end, in turn, of the trestle saddle. The trestle shall be positioned in the test machine with the saddle and column fully extended. A load equal to the trestle's specified capacity shall be applied to the saddle for a time period of approximately tne seconds, as depicted in Figure 3. The load shall then be removed and the trestle visually examined for cracked welds. permanent deformation and other damage. Following the examination, the

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operability of the rack and pawl mechanism shall be verified Evidence of cracked welds, permanent deformation or other damage: failure of the rack and pawl mechanism to operate freely throughout its entire adjustment range: or any tipping, overturning, or movement of the trestle shall be cause for rejection (see 3.5.2). The above test shall be repeated with the saddle and column of the trestle fully retracted (a total of four load tests shall be performed.)

4.5.2 Proof load test-production. The trestle shall be positioned in the testing machine with the saddle and column fully extended. A proof load equal to 1.5 times the trestle's specified capacity shall be applied to the saddle for a time period of approximately ten seconds, as depicted in Figure 4. The proof load shall then be removed and the trestle visually examined for damage. crack ed welds and permanent deformation. Following the examination, the operability of the rack and pawl mechanism shall be verified Evidence of cracked welds or other damage: permanent deformation: or failure of the rack and pawl mechanism to operate freely throughout its entire adjustment range shall be cause for rejection (see 3.5.3).

4.5.3 Functional test-production. After the proof test has been completed. perform the functional test in paragraph 4.4.6 in accordance with the sampling plan.

4.6 Packaging inspection. Packagin inspection shall be conducted before and after packaging to determine compliance with the requirements of Sectin 5.

5. PACI AGING

5.1 Preservation. Preservation shall be Level A or Level C as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The trestles shall be cleaned in accordance with process C-1 and dried in accordance with procedure D-1 or D-4 of MIL-P-116.

5.1.1.2 Preservation. The method of preservation shall conform to Method I or Method III of MIL-P-116. as applicable. All ferrous metal external surfaces of the trestles which have not been painted shall be coated with type P-1 preservative of MIL-P-116 The supporting column of the trestle shall be placed in its innermost position and secured with annealed wire.

5.1.2 Level C Level C preservation shall be as specified in MIL-STD-1190.

5.2 Packing. Packing shall be Level A, Level B, or Level C.

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5.2.1 Level A. The trestles shall be packed in a box conforming to PPP-B-621, class 2. style 2. Alternating rows of trestles shall be inverted to reduce the volume of the shipping container, Blocking and bracing shall conform to MIL-STD-1186. The quantity to be packed in the exterior shipping container shall be governed by the weight limitations of the box specification.

5.2.2 Level B. The trestles shall be packed as for Level A (see 5.2.1) except that shipping containers shall conform to PPP-B-621. class 1.

5.2.3 Level C. Level C packing shall be as specified in MIL-STD-1190.

5.3 Marking. All marking requirements shall be in accordance with MIL-STD-129. Barcode marking is required and shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Trestles covered by this specification are a type of vehicle support stand used for supporting motor vehicles while the vehicles are undergoing repairs and while they are in storage. Storage usually is outside where the trestles are subject to weathering over long periods of time. The number of trestles required per vehicle is dependent upon the weight of the vehicle and the number of axles to be supported.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number , and date of this specification.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).
(c) Load capacity required (see 1.2).
(d) First article when required (see 3.2).
(e) First article inspection when required (see 4.2.1).
(f) Level of preservation, packing, and marking if different (see 5.1, 5.2, and 5.3).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists. and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provision of the Federal Acquisition Regulation. The first article should be a preproduction sample, or a standard

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production item from the contractor's current inventory. The contracting officer should include specific c instructions in all procurement instruments regarding arrangements f or examination, test and approval of the first article.

6.4 Subject term (key word) listing.

Rack
Pawl
Saddle
Base

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - AS
Air Force - 99

Review activities:
Army - ME

User activity:
Army - AT

Preparing activity:
Army - AL

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1: The preload, if used, shall be performed prior to taking measurements.

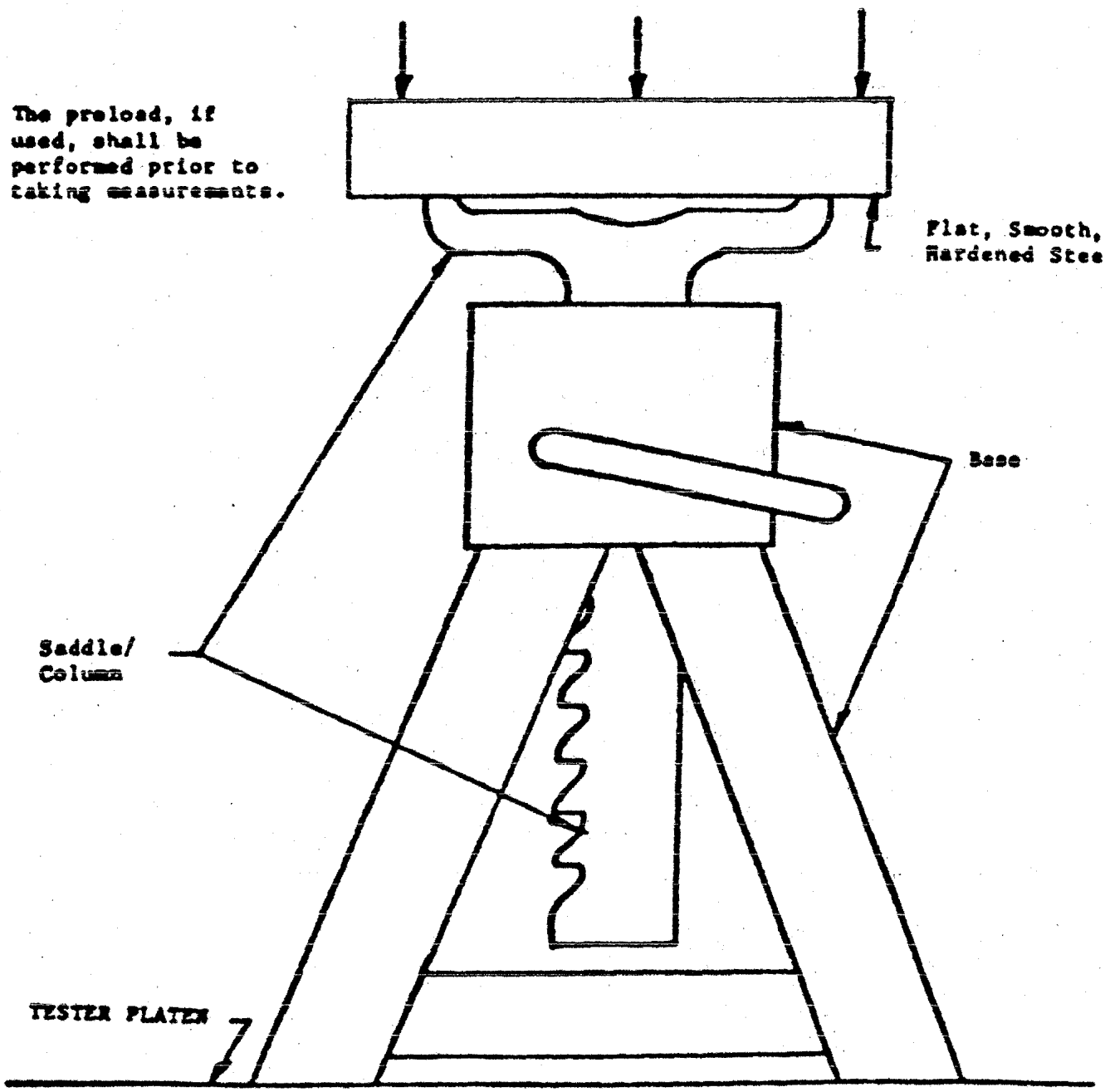


FIGURE 1. Preload (see 3.5.1).

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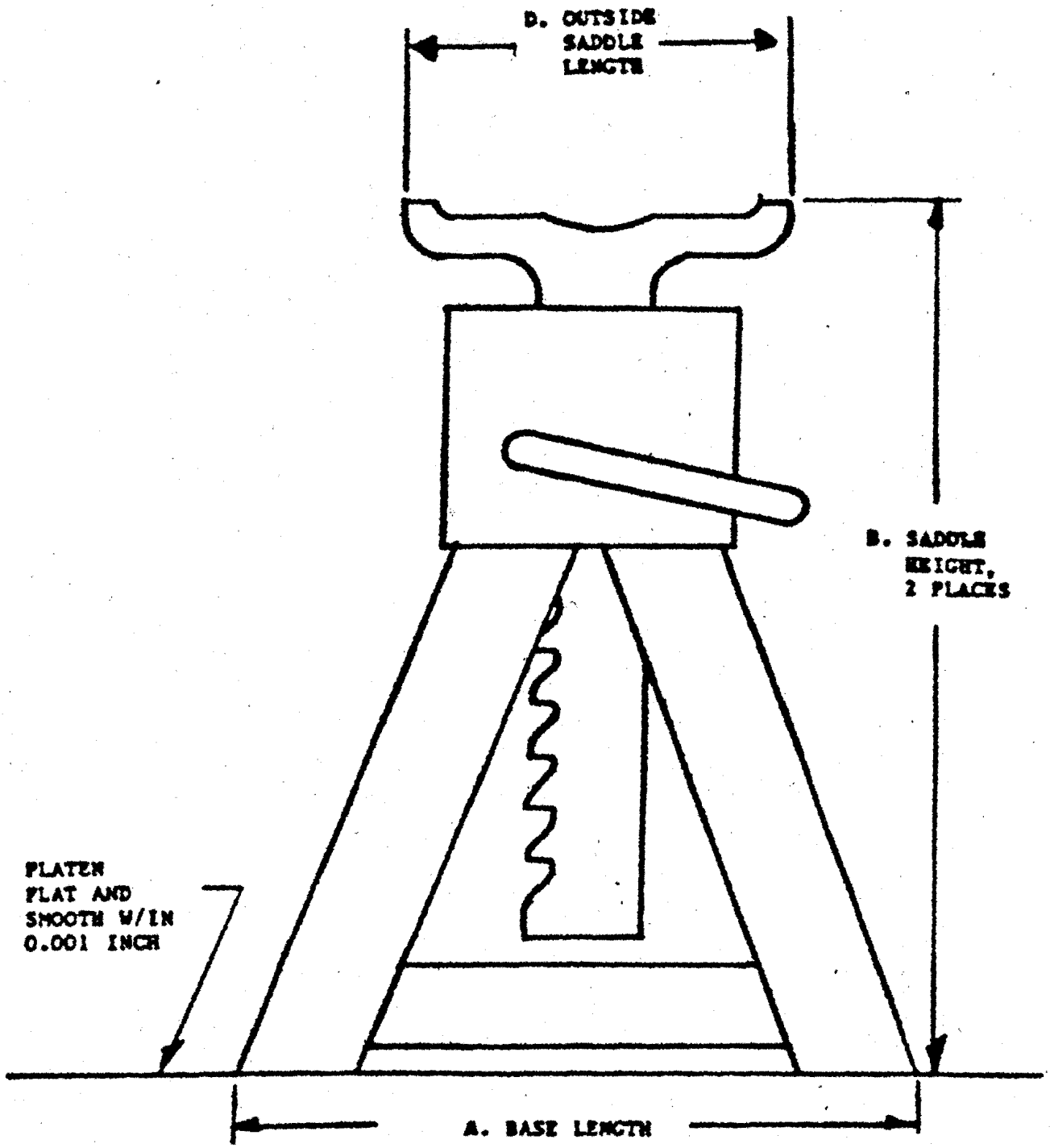


FIGURE 2. Measurements (see 3.5.4).

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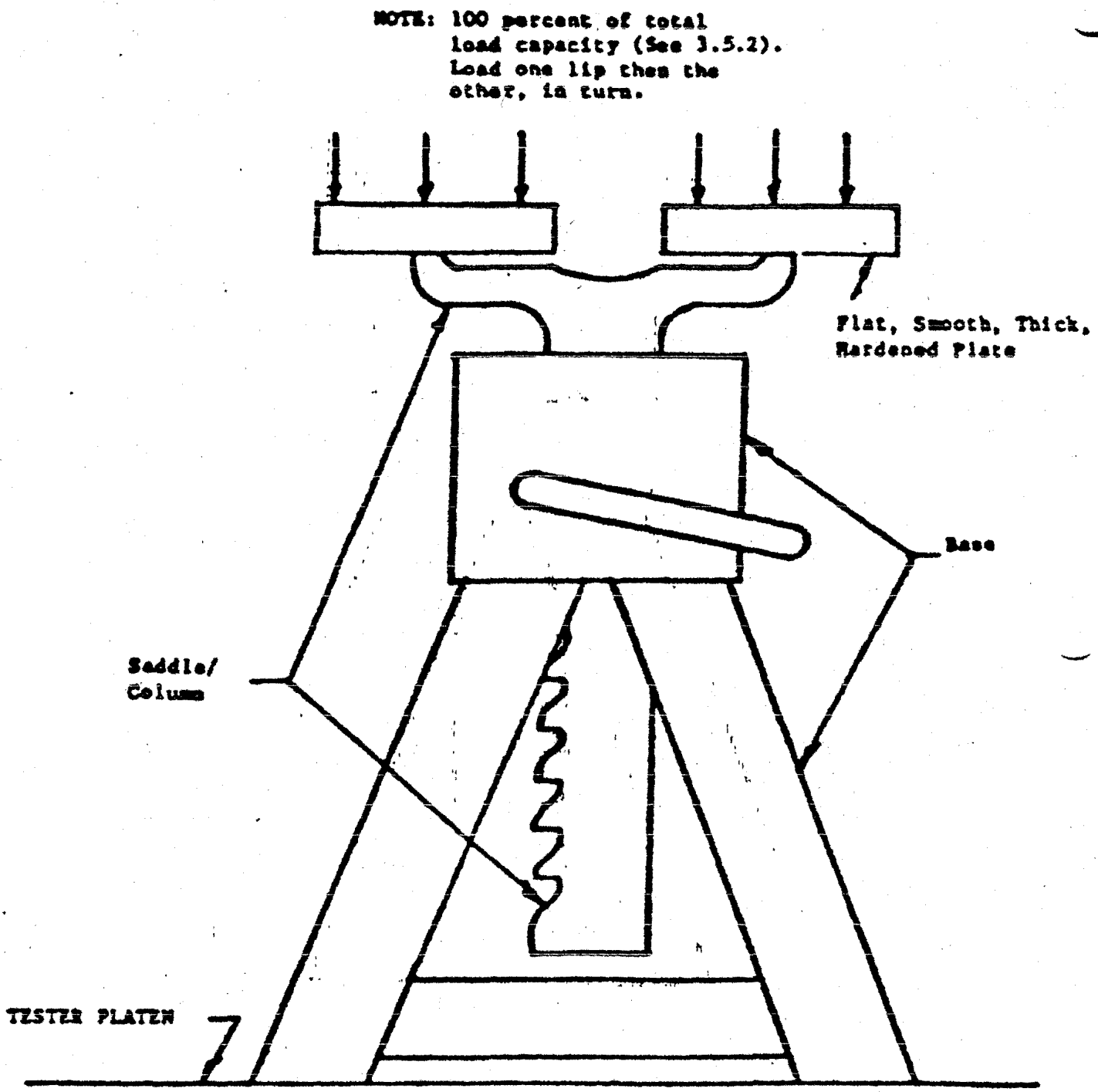


FIGURE 3. Eccentric load.

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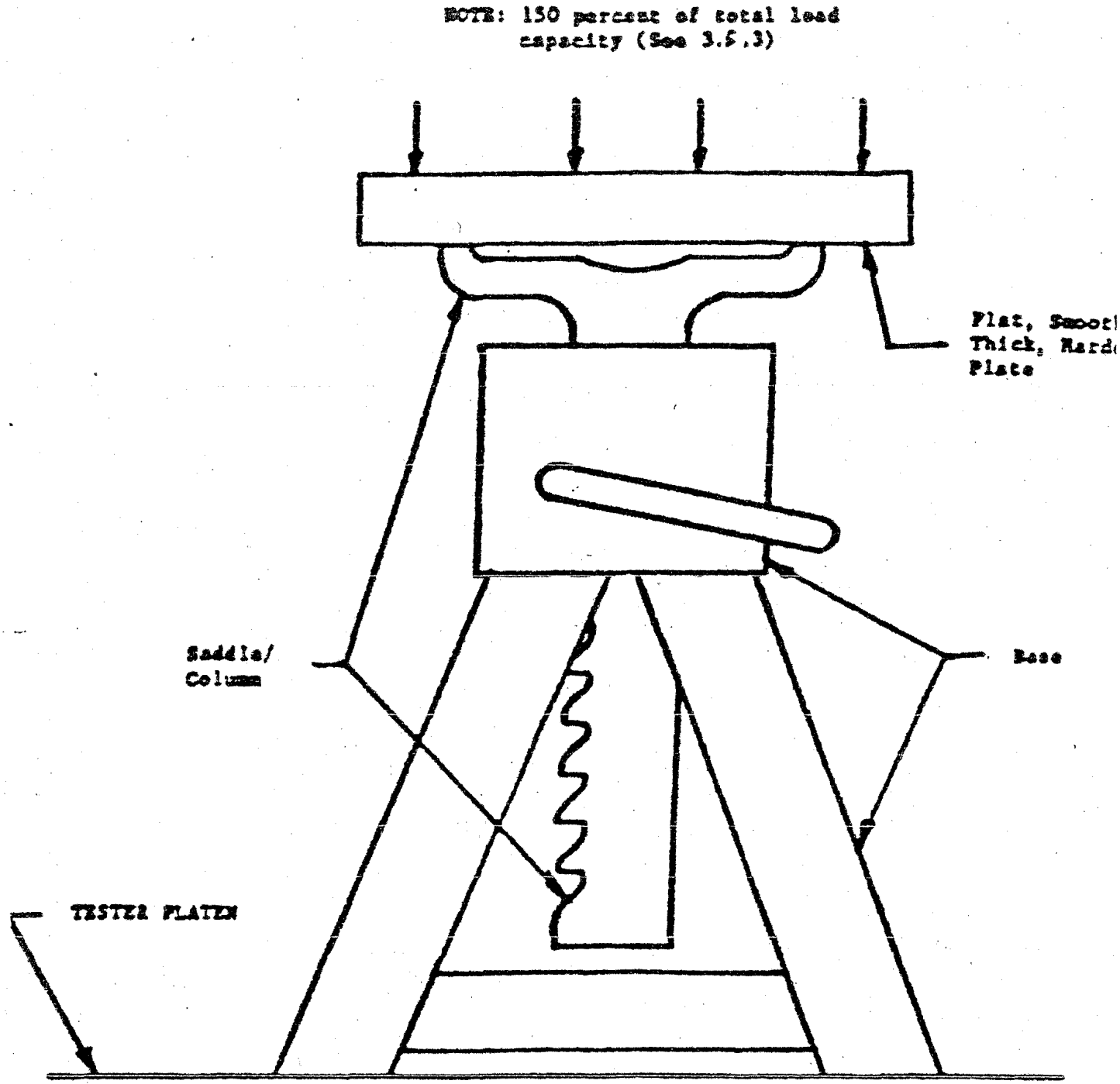


FIGURE 4. Safety factor proof load.

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DOCUMENT IDENTIFIER AND TITLE (Vehicle Support Stands)
MIL-T-14521E, Trestles, Motor Vehicle Maintenance & Storage

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
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